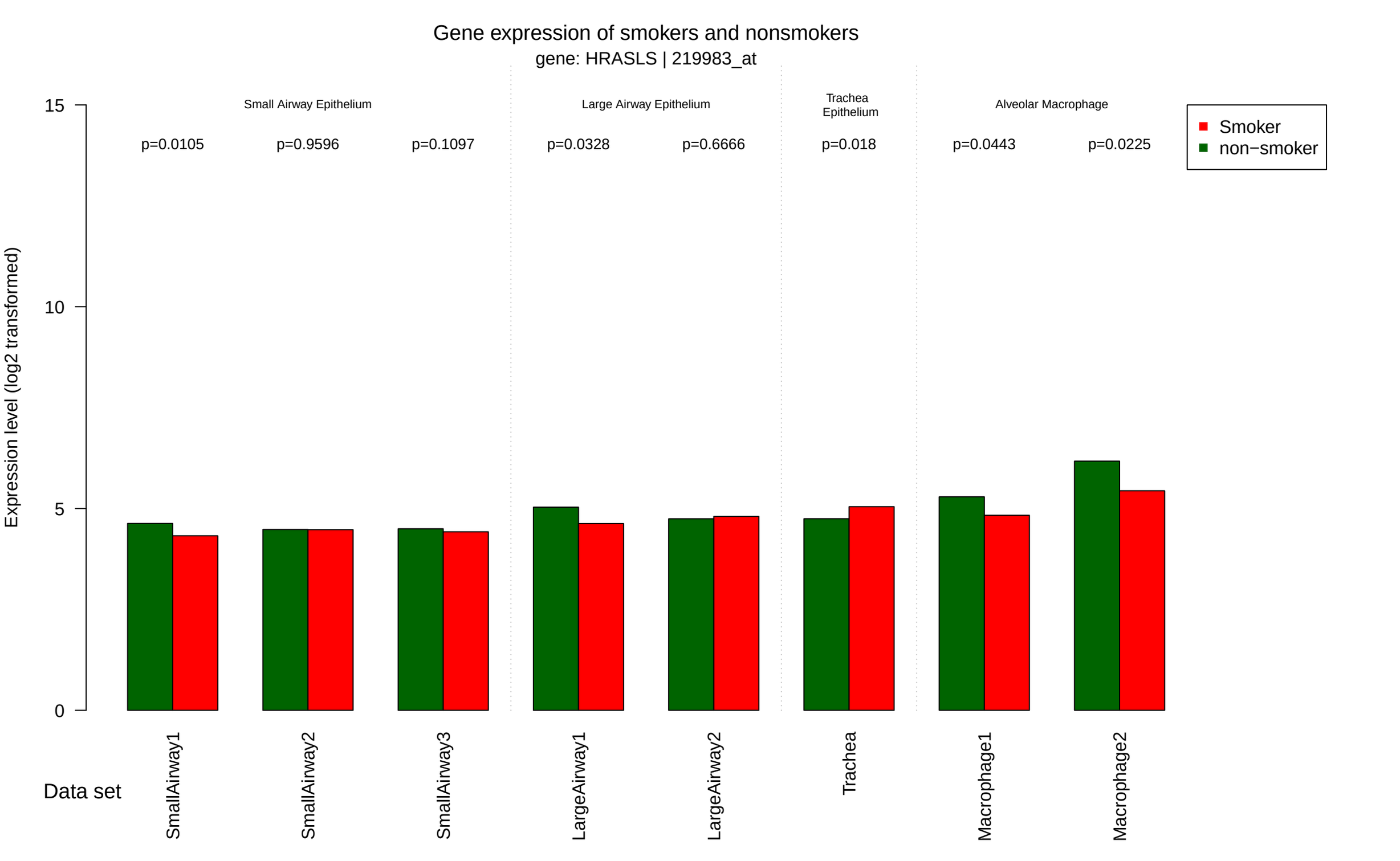
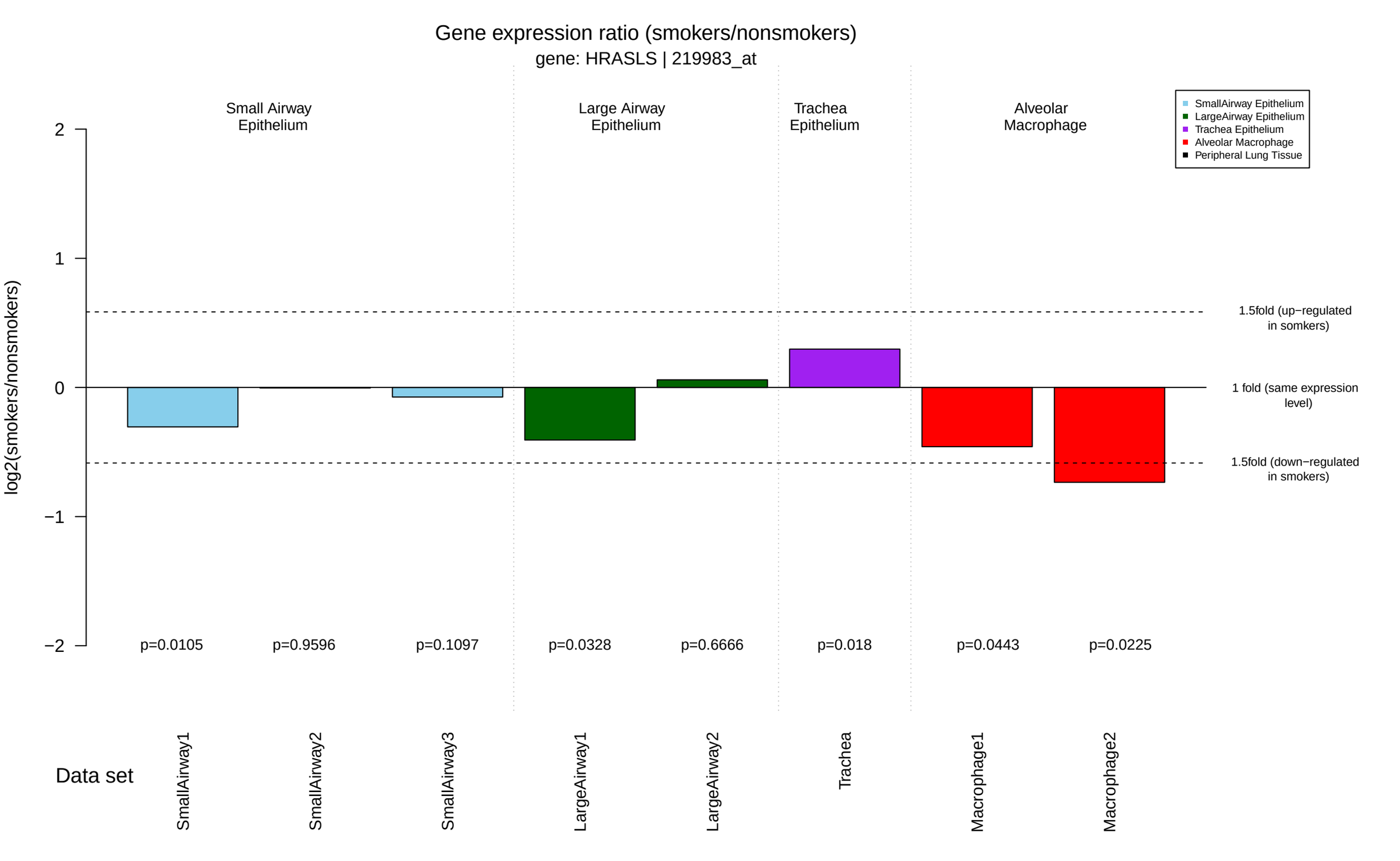
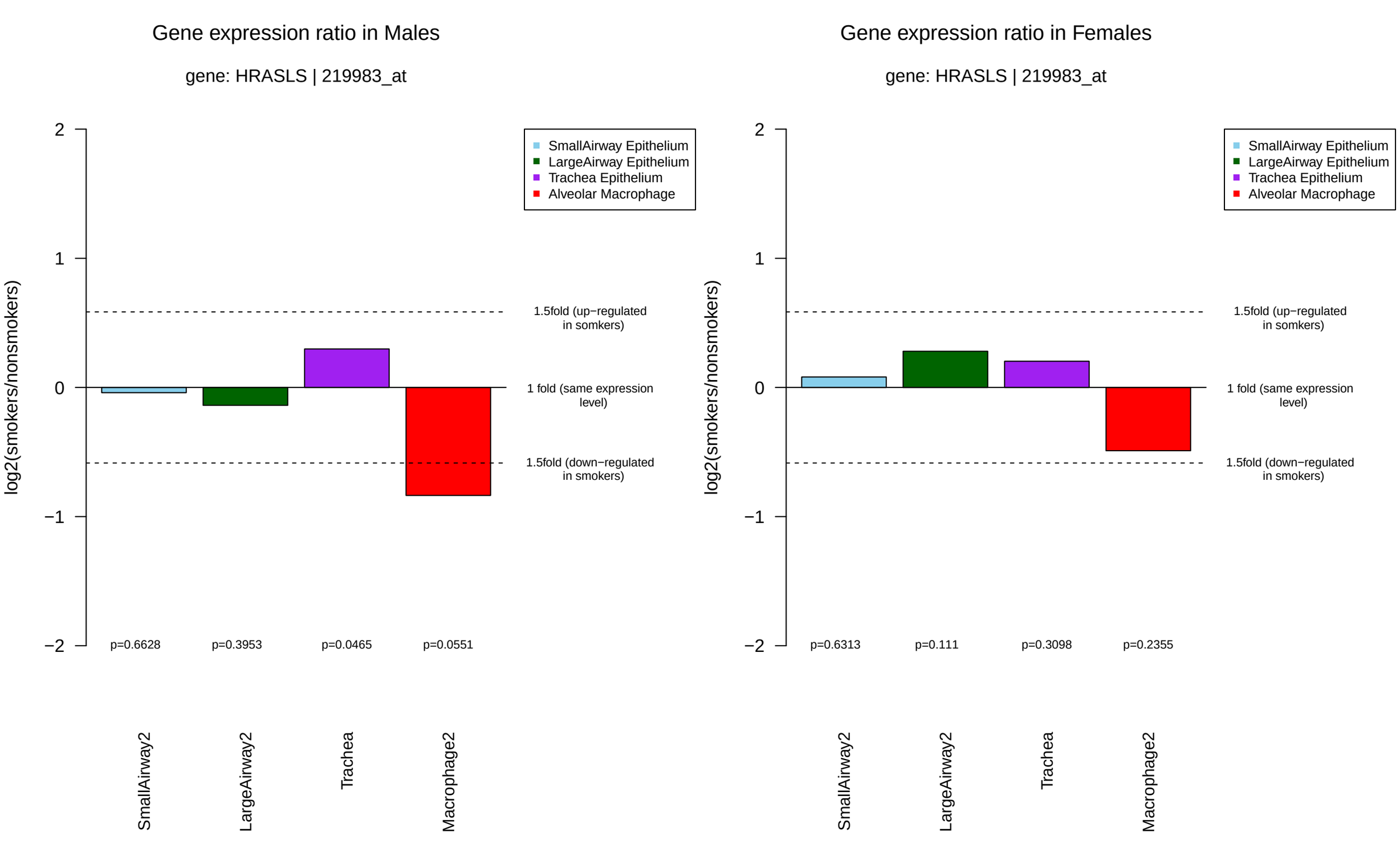
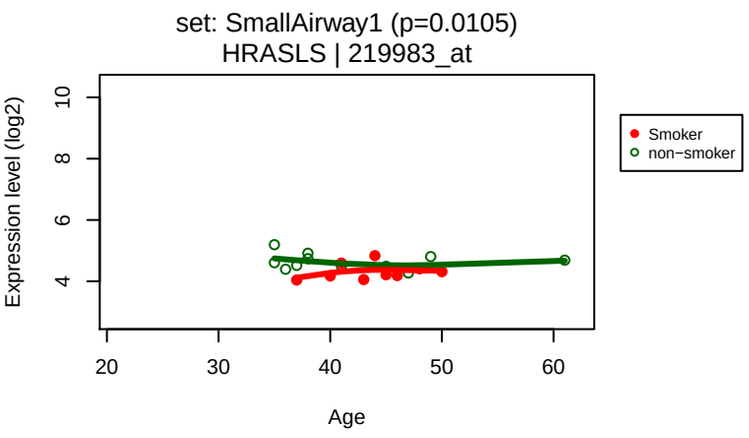
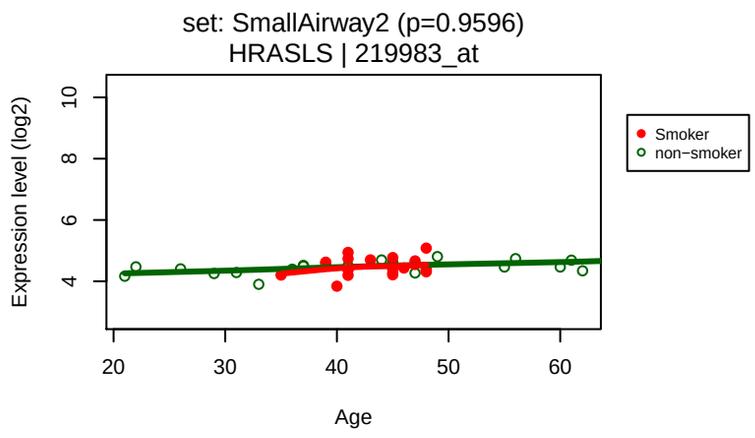
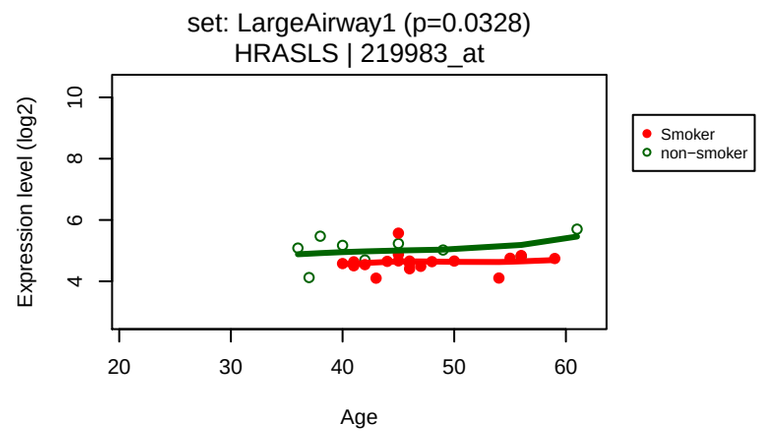

Gene expression of smokers and nonsmokers
gene: HRASLS | 219983_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0105
p=0.9596
p=0.1097
p=0.0328
p=0.6666
p=0.018
p=0.0443
p=0.0225
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

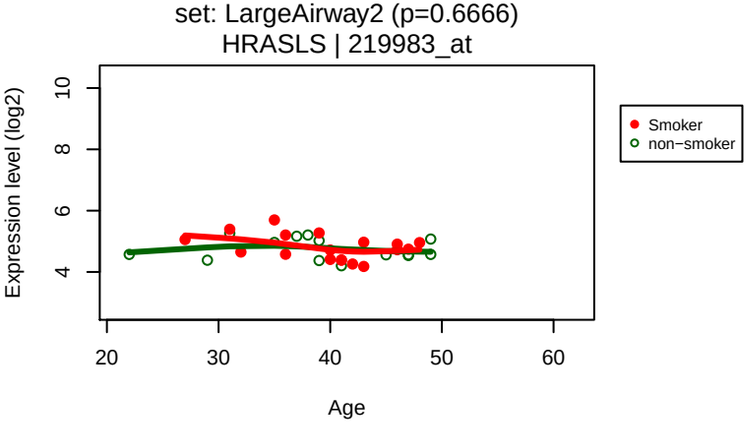

Gene expression ratio (smokers/nonsmokers)
gene: HRASLS | 219983_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0105
p=0.9596
p=0.1097
p=0.0328
p=0.6666
p=0.018
p=0.0443
p=0.0225
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

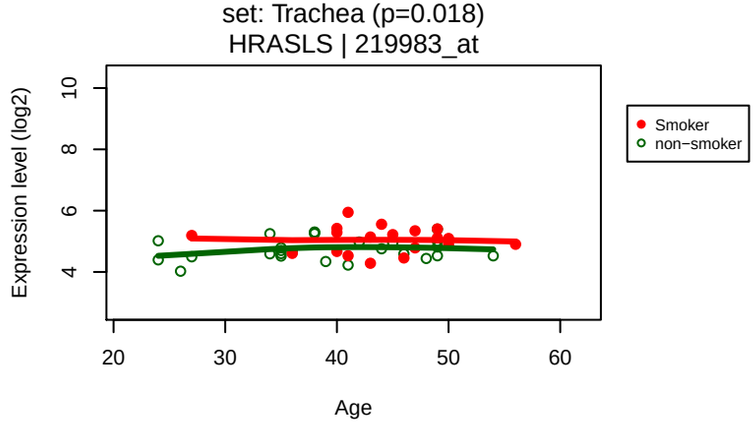

Gene expression ratio in Males
gene: HRASLS | 219983_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6628
p=0.3953
p=0.0465
p=0.0551
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRASLS | 219983_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6313
p=0.111
p=0.3098
p=0.2355
SmallAirway2
LargeAirway2
Trachea
Macrophage2

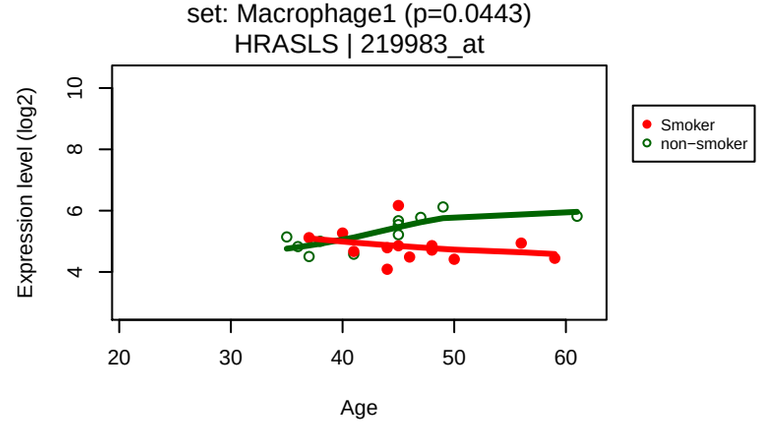

set: SmallAirway1 (p=0.0105)
HRASLS | 219983_at
Expression level (log2)
Age
Smoker
non−smoker

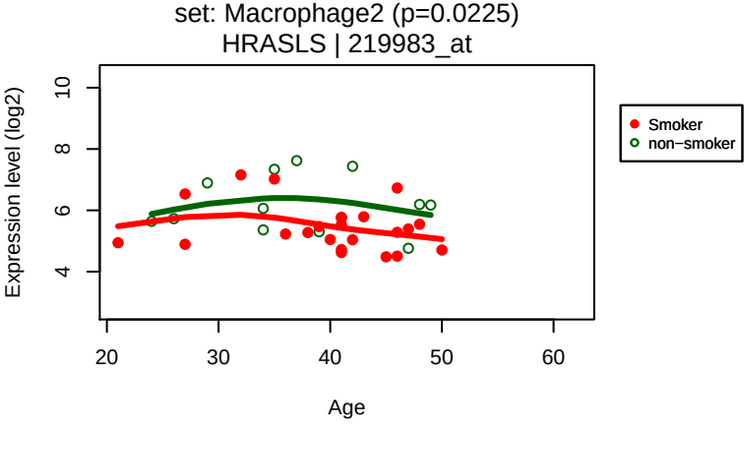

set: SmallAirway2 (p=0.9596)
HRASLS | 219983_at
Expression level (log2)
Age
Smoker
non−smoker

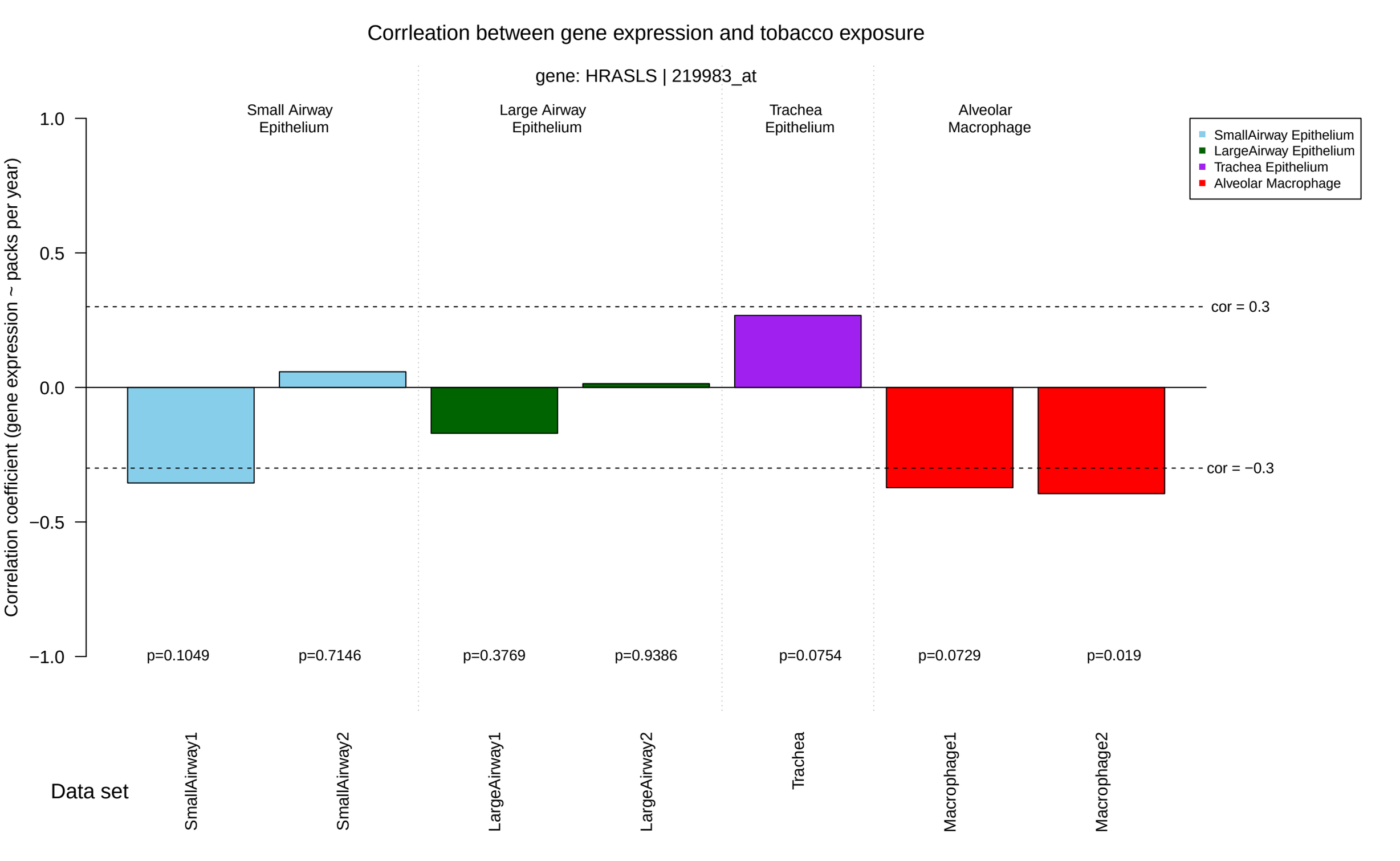

set: LargeAirway1 (p=0.0328)
HRASLS | 219983_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6666)
HRASLS | 219983_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.018)
HRASLS | 219983_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0443)
HRASLS | 219983_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0225)
HRASLS | 219983_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRASLS | 219983_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1049
p=0.7146
p=0.3769
p=0.9386
p=0.0754
p=0.0729
p=0.019
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2